1. Determine the regular expression for all strings containing exactly one ‘a’ over ∑ = {a, b, c}.

Solution:

We have the input alphabets are Σ = {a, b, c}

The objective of the problem is to find out the regular expression for all strings containing exactly one ‘a’.

For this, first find out the regular expression for any strings at all over the given Σ, which does not contain any ‘a’. It is-

(b + c)*

Thus, to write a regular expression which denote all string containing exactly on ‘a’ we simply put the regular expression for any string at all which does not contain any ‘a’ on the both side of ‘a’. Thus, the resultant regular expression is-

(b + c)* a (b + c)*

2. Find the regular expression for the set of all strings over input alphabet ∑ = {a, b} with three consecutive b’s.

Solution:

We have the input alphabets are Σ = {a, b}

According to the problem given, the resultant regular expression represent all strings which have three consecutive b’s (which means, strings always have a substring of b’s of length 3 i.e bbb) over the given input alphabets Σ = {a, b}.

For solving this problem, first find out the regular expression for any string at all over the given input alphabets Σ = {a, b}. It is-

(a + b)*

Thus, to write a regular expression which denote all string with three consecutive b’s we simply put the regular expression for any string at all over the given Σ on the both side of three consecutive b’s, i.e., bbb. Thus, the resultant regular expression is-

(a + b)* bbb (a +b)*

3. Find the regular expression for the set of all strings over {0, 1} beginning with 00.

Solution:

We have the input alphabets are ∑ = {a, b}

Here, the resultant regular expression will denote the set of all strings over the given ∑ = {a, b} which must start with two consecutive 0's i.e. 00.

For this, fist write the regular expression which represent the set of all strings over the given ∑ = {a, b}. It is-

(0 + 1)*

Now, the regular expression which fulfill the requirement of given question can be written as-

00 (0 + 1)*

4. Find the regular expression for the set of all strings over {0, 1} ending with 00 and beginning with 1.

Solution: We have the input alphabets ∑ = {a, b}.

Here, the resultant regular expression will denote, "The set of all strings over the given ∑ which must start with 1 and ends with two consecutive 0's i.e. 00."

To fix this problem we first write the regular expression which represents the set of all strings over the given ∑. It is

(0 + 1)*

Now, the regular expression which fulfill the requirement of given problem can be written as-

1 (0 + 1)* 00

5. Find the regular expression for all strings over input alphabets ∑ = {0, 1} ending in either 010 or 0010.

Solution:

We have the input alphabets ∑ = {0, 1}

Here, the resultant regular expression will denote the set of all strings over the given input alphabets ∑ = {0, 1} which must either ends with the substrings 010 or 0010.

To fix this problem, first write the regular expression which represent the set of all strings over the given input alphabets ∑ = {0, 1}. It is-

(0 + 1)*

Now, the regular expression which fulfill the requirement of the given question can be written as-

(0 + 1)*(010 + 0010)

Note: In all regular expression which are represented in this problem set symbol '+' denotes the 'or' condition, which means we get an acceptable string either first expression or other which are separated by symbol '+'. At some books and these types of problem solution you found '∪' or '|'. Hence each symbol has the same meaning.

6. Find the regular expression for all string containing no more than three a's over input alphabets ∑ = {a, b, c}

Solution:

We have the input alphabets ∑ = {a, b, c}.

Here, the resultant regular expression will denote the set of all string over the given input alphabets ∑ = {a, b, c}, which contains no more than three a's, i.e., it will contains either zero a or one a or two a or three a.

For this, first write the regular expression which denotes zero or one 'a'. It is-

$(\lambda + a)$
Now, we can write either zero 'a' or one 'a' or two 'a' or three 'a' as –

$(\lambda + a)\ (\lambda + a)\ (\lambda + a)$

Now, since we have given $\Sigma = \{a, b, c\}$, as per the problem given the resultant regular expression denotes set of all strings containing no more than three a's over the given input alphabets $\Sigma = \{a, b, c\}$.

Thus, we must have to allow the arbitrary regular expression which represent the string not containing a's over the given input alphabets $\Sigma = \{a, b, c\}$ at the place which marked by 'X' (Assumed), as under –

$X\ (\lambda + a)\ X\ (\lambda + a)\ X\ (\lambda + a)\ X$

Here, the regular expression for the place X can be written as-

$(b + c)^*$

i.e., the regular expression over the given input alphabets $\Sigma = \{a, b, c\}$ which does not contain any 'a'.

Thus, putting this regular expression at the place marked by X's in the form of resultant regular expression given above, we will get;

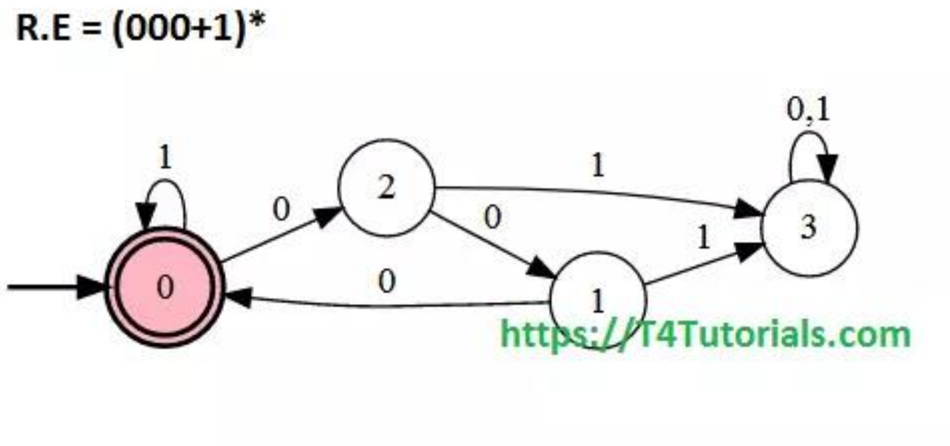
$(b + c)^*\ (\lambda + a)\ (b + c)^*\ (\lambda + a)\ (b + c)^*\ (\lambda + a)\ (b + c)^*$

This is the required resultant regular expression.

7. Regular Expression for no 0 or many triples of 0's with no 1 or many 1 in the strings

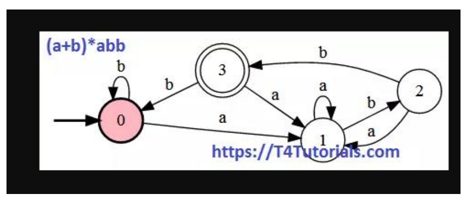
R.E = (000+1)*

DFA for R.E = (000+1)*



7. **A regular expression for ending with abb**

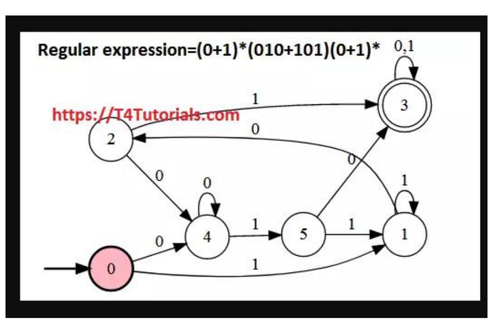
Regular expression=(a+b)*abb



8. Regular expression for all strings having 010 or 101

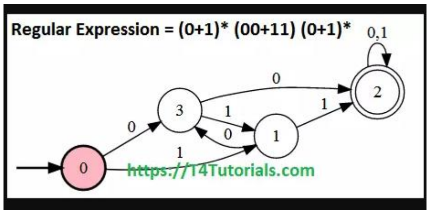
Let us see the Regular expression for all strings having 010 or 101 defined over {0,1}

Regular expression= (0+1)*(010+101)(0+1)*



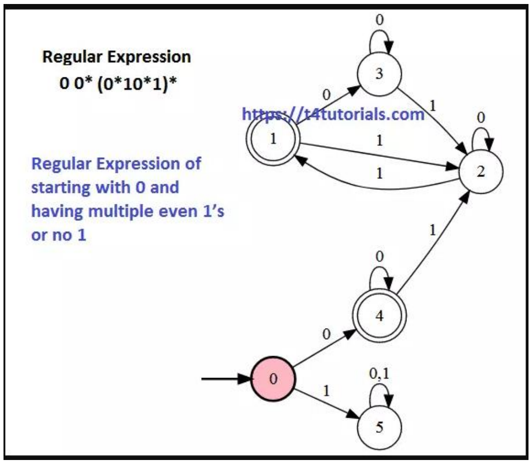
9. Regular Expression for strings having at least one double 0 or double 1

Regular Expression = (0+1)* (00+11) (0+1)*



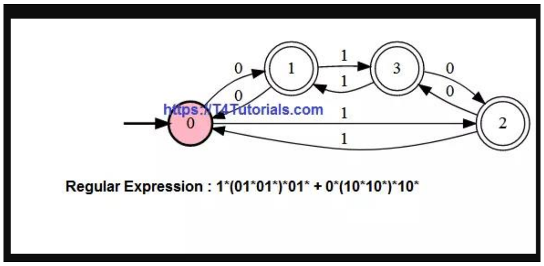
10. Regular Expression of starting with 0 and having multiple even 1’s or no 1

Regular Expression = 0 0*(10*10*)*



11. **Regular Expression for an odd number of 0’s or an odd number of 1’s in the strings**

Regular Expression : 1*(01*01*)*01* + 0*(10*10*)*10*

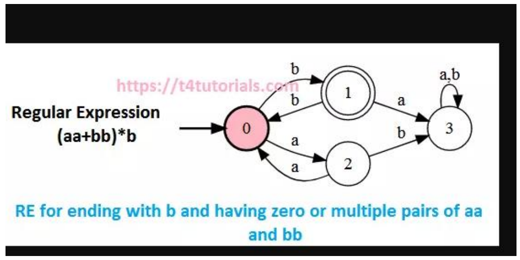


12. Regular Expression for having strings of multiple double 1’s or null

Regular Expression: (11) *

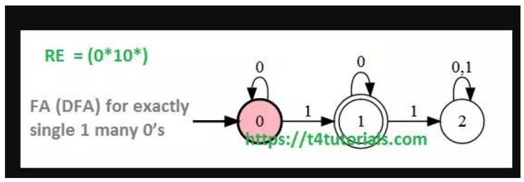
13. RE for ending with b and having zero or multiple sets of aa and bb

Regular Expression = (aa+bb)*b



14. RE for exactly single 1 many 0's

RE = (0*10*)



15. Regular expression for the language of all consecutive even length a's